

Degradation analysis of perovskite solar cells doped with MABr₃ via electrochemical impedance

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ABSTRACT

In this study, perovskite solar cells with a structure Cs₇FA_{1-x}MA_xCS_{0.05}Pb(I_{1.05-x}Br_x)₃, with X = 0 (no doping was added to the base structure CsFAPbI₃) and X = 0.085 (CsFAPbI₃ was doped with 8.5% MABr₃) were prepared and were subjected to a 60 days ageing process (15 days under vacuum conditions followed by 45 days under room conditions) to study their stability. J-V curves showed that after the 60 days, the cells with no MABr₃ only had 50.32% of the original efficiency remaining, while the samples with 8.5% MABr₃ had 92.89% of the original efficiency, being the J_{sc} the main parameter that was affected during the ageing period. IPCE measurements suggest the formation of Cs segregates, that maintain or even increase the J_{sc}. According to electrochemical impedance studies, these segregates could be promoted by the lower ionic movement resistance in the 8.5% MABr₃ samples, which may allow to re-arrange the atoms of the crystalline structure in a benefic way. This work is of great importance because it helps to understand how the doping affects the degradation processes, while showing how Impedance Spectroscopy can be used to characterize the changes in the electrical and ionic dynamics of the solar cells.

1. Introduction

Perovskite solar cells (PSCs) have found a rise in interest in the last decade thanks to their excellent photovoltaic properties and low preparation cost [1,5,15]. The highest reported efficiency for PSCs has been 25.5%, making them competitive against the commercial solar cells. However, PSCs present some problems that makes them not yet viable for commercialization. Their main challenge to overcome is their high sensibility against the different environmental factors (such as heat, humidity, oxygen and light), which produces low long-term stability and durability on the perovskite based solar cells [30,62]. For this reason, stability has been a major focus in the scientific community, which has led to the development of various approximations to solve degradation issues. One of the methods to improve stability is by modifying the structure of the different sites (A, B and X) in the perovskite structure via doping [12]. Doping the perovskite with ions that are geometrically stable and fit in the A site has proven to be the most effective doping mechanism [14,42,61], like the use of Rubidium (Rb) to increase the

thermal stability of the perovskite layer [39,55,64]. Another method consists in the use of big-sized organic molecules to separate the perovskite in low dimensioned layers to create hybrid 2D/3D PSCs that allows the reduction of surface defects and increases the hydrophobicity of the perovskite layer, enhancing stability [20,22,28]. While these and other methods have been beneficial on the stability of the perovskites, there has not been enough research to understand why these techniques have such positive impact.

Although J-V measurements provide the basic data to showcase the overall performance of a solar cell, and its changes across time, it doesn't give enough information regarding the causes of said performance, much less of the causes in deterioration. For that reason, additional techniques and measurements are required. Impedance Spectroscopy is a technique that helps in the understanding of different internal mechanisms and processes and how they affect the charge transfer in the different layers and interfaces of the cell structure. This not only includes the electrical and electrochemical processes of perovskite solar cells [36,46], but also of various types of solar technologies such as

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organic solar cells, dye sensitized solar cells, silicon solar cells, etc [8,17,32,35,38,53]. Due to the robust data this technique provides, Impedance Spectroscopy is not only employed to analyze the internal phenomena of the freshly prepared cells, but also to quantify the changes of the internal mechanisms across aging periods to better understand the reasons of degradation of the cells [2,10,52,56].

In this work, the effects of different storage conditions towards CsFAPbI₃ PSCs degradation were studied, as well as the effects of using methylammonium bromide as a dopant in the stability of the devices. Both doped and undoped PSCs were prepared and subjected to an ageing process under the dark, and during this process both type of samples were characterized via J-V curves to compare their performance across the ageing period, which shows that the undoped PSCs at Day 45 retained 33% of the original efficiency, while the doped PSCs retained 82%. The IPCE and Impedance Spectroscopy studies suggests that the main reason for the PSCs doped with MABr to retain better efficiency is due to the formation of benefic photoactive segregates in said cells, situation that didn't happen in the undoped cells.

2. Experimental methods

2.1. Materials

Patterned fluor tin oxide (FTO)-coated glass substrates, lead iodide (PbI₂, TCI, 99.99%), lead bromide (PbBr₂, TCI, 99.99%), formamidinium iodide (FAI, dyenamo, >99%), methylammonium iodide (MAI, dyenamo, >98%), cesium iodide (CsI, Aldrich, 99.9%), spiro-MeOTAD (Lumtec, LT-S922, 99%), N,N-dimethylformamide (DMF, Aldrich, anhydrous 99.8%), dimethyl sulfoxide (DMSO, Aldrich, 99.9%), Isopropanol (IPA, Aldrich, anhydrous 99.5%), lithium salt (LiTFSI, Aldrich, 99.95%), acetone, acetonitrile, ethanol, and acetylacetone (CH₃COCH₂COCH₃, Honeywell, ≥99.5%) were used as purchased.

2.2. Device fabrication

Substrates of glass coated with FTO previously etched were cleaned by ultrasonication in detergent, deionized water, acetone, and IPA. Compact TiO₂ was deposited by spray pyrolysis. Following a mesoporous TiO₂ layer is deposited via spin coating at 2000 rpm for 10 s and annealed at 450 °C for 2 h. Then the samples were subjected to a passivation process using a lithium salt: LiTFSI in acetonitrile 10 mg/mL, which was spin coated at 1000 rpm for 10 s and heated similarly to the *meso*-TiO₂ (450 °C annealing for 2 h). A solvent composed of DMF: DMSO (4:1 V:V) was used to dissolve mother solutions of PbI₂ and PbBr₂ at 1.5 M concentration, to be added FAI and MABr at 1.24 M concentration, respectively. Additionally, CsI was dissolved at 1.5 M in DMSO to be used as a dopant, then the solutions were mixed to obtain the desired stoichiometries accord to the experiments. The obtained solutions were spin coated on the substrates according to previous works [48]. Next spiro-MeOTAD was deposited via spin-coating at 1800 rpm for 30 s. Finally, using thermal evaporation, a gold film with a 80 nm thickness was thermally deposited as a metal electrode. The final device presents the general structure Glass/FTO/TiO₂/Perovskite/Spiro-MeOTAD/Au, as shown in Fig. 1a. Two perovskite compositions were tested: CsFAPbI₃ (denoted as 0% MaBr) and CsFA_{0.915}MA_{0.085}Pb(I_{0.915}Br_{0.085})₃ (denoted as 8.5% MABr). 20 samples were prepared and characterized for each composition. The 8.5% MABr concentration was selected considering previous work where this concentration yielded the best performance and stability [48].

2.3. Characterization

X-Ray diffraction XRD was done with a Bruker 2D Phaser with a x-ray source of 1.54 Å. UV–Visible absorption were carried using an Agilent Technologies Spectrophotometer (Gary 5000).

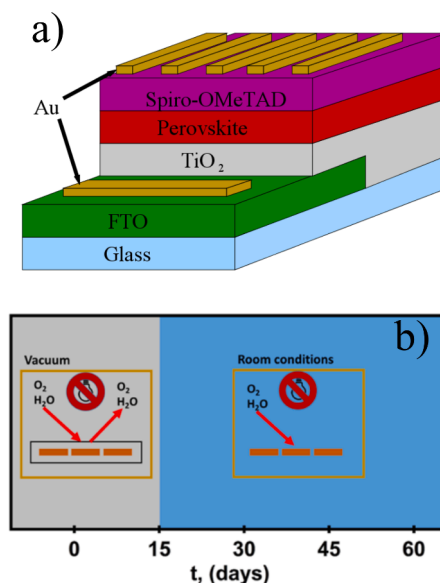


Fig. 1. A) schematic of the perovskite solar cell structure. b) ageing process for the perovskite solar cells. in the first 15 days the cells were subjected to a vacuum environment, and on the following 45 days, the cells were submitted to room conditions. The cells were stored in the dark in the whole process.

Photoelectrochemical measurements (open-circuit voltage, J-V curves, IPCE and impedance spectroscopy measurements) were carried out using a GAMRY Instruments Reference 600 potentiostat. Current density–voltage curves (J-V curves) were measured from open circuit to short-circuit (reverse scan), with a scan rate of 100 mV/s and a step size of 0.1 mV under a 100 mW/cm² AM1.5G (1 sun) illumination from a VERASOL 3A Solar Simulator. Incident photon to current conversion efficiency (IPCE) measurements were carried on with an 66,021 Oriel Xenon lamp and 77,200 Oriel monochromator making wavelength scans from 350 to 850 nm. Impedance spectroscopy measurements were performed at open-circuit conditions under different irradiances, applying a small voltage perturbation (10 mV), at frequencies that range from 10⁵ to 10⁻¹ Hz. All light sources were calibrated with a Newport 91150 V Calibration Si cell with quartz window.

2.4. Aging

With the objective of identifying how the samples deteriorate under different conditions, these were subjected to inert conditions and to non-controlled conditions with humidity and oxygen. First, the solar cell devices were stored under dark conditions for 15 days in vacuum conditions, followed by 45 days at room temperature, with relative humidity ≥ 60%, as shown in Fig. 1b. Temperature and humidity in the second stage were only monitored, according to the International Summit on Organic PV Stability dark storage (ISOS-D1) protocol [24].

3. Results

3.1. Morphological and optical characterization

The proper doping of the 8.5% MABr samples was verified by XRD and absorbance. The XRD spectra (Fig. S1), shown that the undoped samples present the characteristic peaks of the CsFAPbI₃ perovskite crystalline lattice, while the 8.5% MABr doped samples present the same peaks with a shift, indicating a change in the lattice parameters due to the incorporation of MABr [26,47]. No additional peaks were detected, indicating that there is no phase segregation, further confirming the appropriate incorporation of the doping in the perovskite layer.

Fig. S2a shows the absorption spectra for both compositions, the 0%

MABr samples presented a shoulder roughly at 775 nm, corresponding to a band gap (E_g) of 1.53 eV (see Fig. S2b). Meanwhile, the 8.5% MABr samples have a slightly shorter absorption range (reaching 760 nm), which correspond to 1.58 eV. This difference in both spectra is consequence of the doping of the photogeneration layer [21,40,45].

3.2. J-V curves

The photoelectrochemical responses of the samples were monitored each 15 days starting from the day they were prepared (day 0). Mean photovoltaic parameters and champion performances of both compositions are reported in Table S1. At day 0 (Fig. 2), it is observed that the fresh samples with 0% MABr have a champion photoconversion efficiency (η) of 13.9% ($J_{sc} = 21.3 \text{ mA/cm}^2$, $V_{oc} = 0.977 \text{ V}$ and $FF = 66.9\%$) with a mean η of 12.1 ± 1.9 , indicating a similar performance than the reported for this composition in the literature [25,27]. On the other hand, the samples with 8.5% of MABr had a slightly lower performance with champion efficiency of 12.0% ($J_{sc} = 19.8 \text{ mA/cm}^2$, $V_{oc} = 0.914 \text{ V}$ and $FF = 66.2\%$) and a mean η of $10.3 \pm 1.2\%$. This performance reduction with the addition of MABr is consistent with previous reports, where it has been stated that the MABr introduction in FAPbI based perovskites increases the band gap, resulting in a reduction of the photogeneration processes and finally limiting the photocurrent in a similar way than the observed in our samples [48].

Analyzing statistical temporal evolution of the photovoltaic parameters of the samples (Fig. 3 and Table S1) it is observed that the 0% MABr cells stored at vacuum conditions for 15 days have almost the same η than at the day they were fabricated, with a performance diminished by a small reduction of J_{sc} compensated by a small increase in both V_{oc} and FF . This suggests a very slow degradation process. However, when these samples were stored at high humidity conditions, a considerable performance loss is observed, reducing 2/3 of its efficiency in 15 days ($\eta = 5.47 \pm 3.26$ at day 30 of the experiment) followed by an almost linear efficiency reduction for the next measurements, being the lost in J_{sc} the principal source of performance decrease.

On the contrary, 8.5% MABr cells presented a considerable efficiency loss when stored in vacuum conditions (from $10.25 \pm 1.24\%$ to 2.67 ± 1.99 after 15 days) followed by a recover in their performance when stored at high humidity conditions, reaching almost the same η than fresh samples at day 45 (mean = 8.45 ± 2.79 , champion 11.12%) to finally decay at day 60. In a similar way than the 0% MABr samples, the changes in performance in the 8.5% MABr samples are principally due to changes in J_{sc} with a small contribution of the V_{oc} that slowly deteriorates and a more prominent contribution of the FF that keeps almost constant until day 45 and suddenly drops at day 60 (from 62.7 ± 12 to $36.4 \pm 4.9\%$).

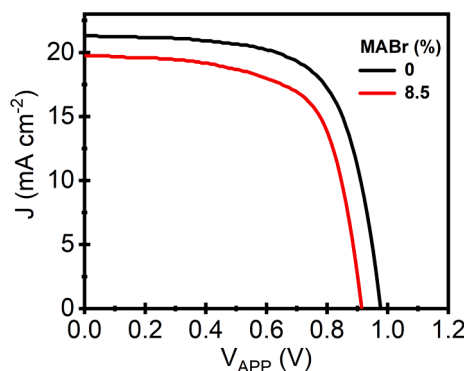


Fig. 2. J-V Curve comparison of the champion cell of both compositions of the perovskite layer at Day 0.

3.3. IPCEs

The identical behavior shared between efficiency and J_{sc} gives the idea that the degradation and recovery of the perovskite layer affect the photogeneration and/or charge recollection mechanism inside the perovskite, as those processes are the main delimiters of J_{sc} [6,16,19], while leakage and recombination process, mostly related FF and V_{oc} [13,18], are less modified. To have a better understanding of the photogeneration mechanism, IPCE measurements for both compositions were done each 15 days.

Fig. 4 showcases the external quantum efficiency EQE (continuous line), and integrated J_{sc} (dashed line) temporal evolution of both compositions. On the 0% MABr cell, at day 15, there is a 90% EQE peak around 370 nm wavelength. This peak in perovskites is related to the presence of PbI_2 phases, either produced by degradation, or PbI_2 molecules that didn't react in the precursor solution [23,41,43,50]. At day 30, it is observed how the EQE has a slight decrease, mainly observed at the decreasing peak around 400 nm, and the disappearance of the 700 nm peak. Subsequent days present a further decrease on the EQE. This tendency suggests the degradation of perovskite (peak at 700 nm), leaving mostly PbI_2 present (400 nm) [31,44,61]. This is reflected on the continuously decreasing J_{sc} .

The 8.5% MABr cell, at day 15, presents an EQE above 50% in the wavelength range of 400–780 nm, which is consistent to the E_g of 1.54 eV [25,34]. The EQE spectrum shows a separation in two shoulders, one at 400–650 nm, and the other at 650–780 nm (58% and 54%, respectively). The second shoulder suggests the presence of photoactive Cs phases, probably due to the degradation process [29,34], which is contrary in the samples without MABr, where no such peaks appeared, meaning that they didn't present segregation of Cs phases, possibly due to a stronger bonding force in FAI than in MABr [11,63]. At day 30, both shoulders are more defined, with an increase in the EQE (79% and 84% for the left and right one, respectively). This suggests a higher segregation of the photoactive Cs phases. The notorious increase in the EQE is the principal responsible of the improvement of J_{sc} . At Day 45, there is a slight decrease in the rightmost shoulder, and on Day 60, both shoulders decrease considerably, which make the EQE identical to the one at Day 15. The disappearance of these shoulders suggests the recrystallization of the perovskite without the Cs segregates.

3.4. Impedance analyses

Although the η is strongly influenced by J_{sc} , there exist significant variations on the FF on both samples with the ageing, suggesting changes in the charge carriers, recombination, leakage, and transport processes [7,33]. On the other hand, V_{oc} remains almost constant at the time that J_{sc} decreases. Since the V_{oc} is determined by the balance between photogeneration and recombination processes [7,33], and photogeneration is strongly modified during the aging, the V_{oc} behavior could indicate a reduction in the recombination processes.

In summary, the correlations between the charge kinetics and the device degradation process are poorly comprehended. To gain understanding of these correlations, impedance measurements at open circuit conditions under several irradiances were performed. Fig. 5a shows characteristic Nyquist plots of the 0% MABr samples at day 15. Here it is observed that all impedance spectra have two well-defined arcs that exponentially decrease with the irradiance, all studied samples show similar impedance trends in good correspondence with the reported in the literature [48,57,59,60].

Equivalent circuit shown in Fig. 5b was used to analyze the impedance curves. This circuit is one of the most extended models [57,60] and provide results that can be correlated with other models including the circuit with elements arranged in series or as alternative model to a circuit using transistors [48]. In this circuit, R_s is the series resistance and represents the resistance to the charge carriers flux due to contacts and wires [37,47,57,59,60]. C_g is the geometrical capacitance (related

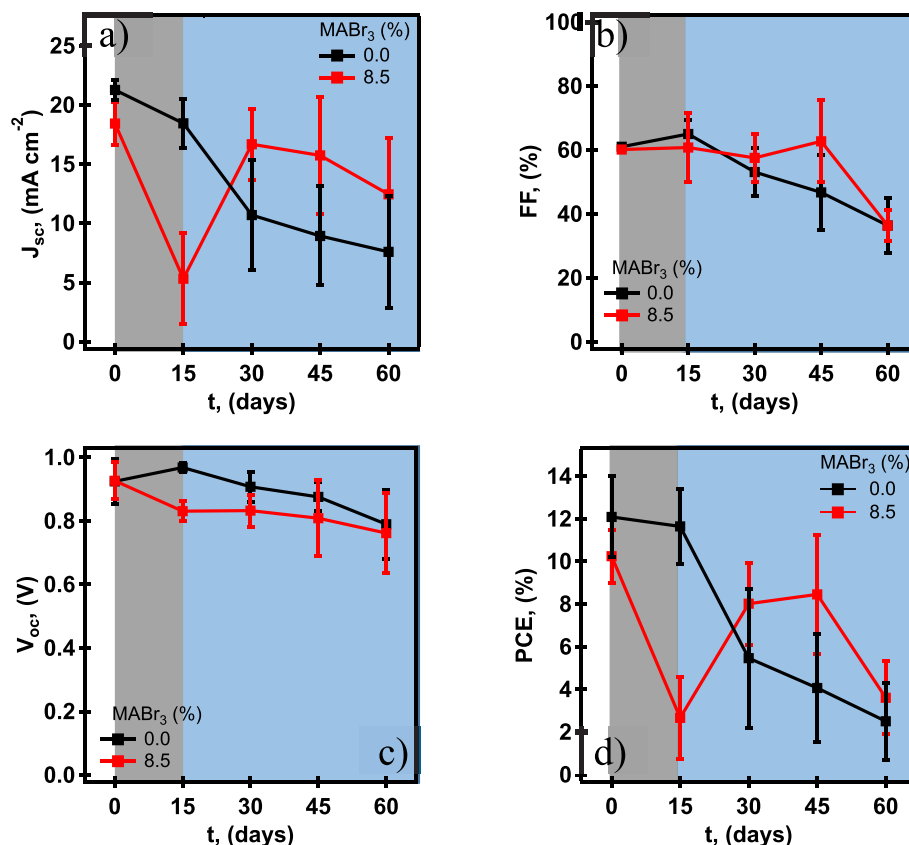


Fig. 3. Photovoltaic parameter statistical temporal evolution calculated from the analysis of 20 samples from each composition.

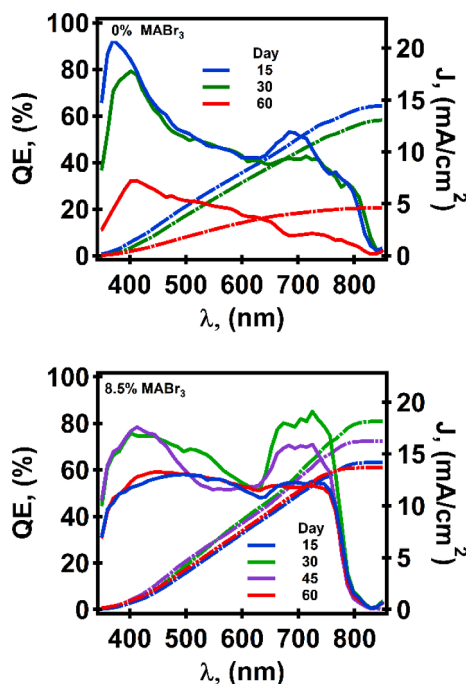


Fig. 4. Temporal comparison of EQE and integrated J_{sc} of both compositions.

to the dielectric properties of the perovskite) [4,3,59,60].

R_{dr} and C_{dr} are the dielectric relaxation (dr) resistance and capacitance of the perovskite respectively. These circuit elements have not been completely related with a single physical process but influenced by several ones. R_{dr} is associated to the ion mobility and the ion vacancy

densities [37,48]. Meanwhile, C_{dr} has been previously studied and associated with the charge accumulation at the interfaces, being the principal cause of JV PSC hysteresis [48,58], although different ion parameters affect its value [48].

R_1 represent the contribution of three resistances arranged in series and parallel as shown in the top branch of Fig. 5c that are: The effective recombination resistance (R_{rec}), related with the recombination processes [37,47,60]. The transport resistance (R_t), which represents the resistance of the perovskite and selective layers to the charge carriers' flux [37,47,57,60]. The Shunt resistance (R_{sh}) related with the charge leakage processes [47,60]. Although the impedance spectroscopy can't differentiate these three elements, as they are associated to similar time constants, it is possible to choose light and voltage conditions that inhibit the contribution of a selected resistance, making possible to analyze their individual behavior in a similar way that are studied from the derivate of the JV curves [47].

3.4.1. Electronic components

Fig. 6a shows the temporal evolution of the series resistances for both compositions. Overall, for both samples and across all the aging process, R_s is below $0.5 \Omega \text{ cm}^2$. These low values of R_s indicates that the electric contact is of good quality for both compositions, being slightly better for 0% MABr samples. For both compositions, R_s presents a small increase as time passes on. This suggests a minor deterioration in the quality in the contact between layers.

Fig. 6b shows the value of R_1 at different voltages for the samples after 45 days of aging. For both samples, the resistances show two types of behavior: at low voltages (below 0.6 V), the resistances have an almost constant value, and at high voltages (above 0.6 V), the resistances present an exponential decrease, similar trends were observed for both samples at all ageing times (see Fig. S3). These two behaviors indicates that R_1 contains R_{sh} and R_{rec} in parallel, as shown in Fig. 5c. As

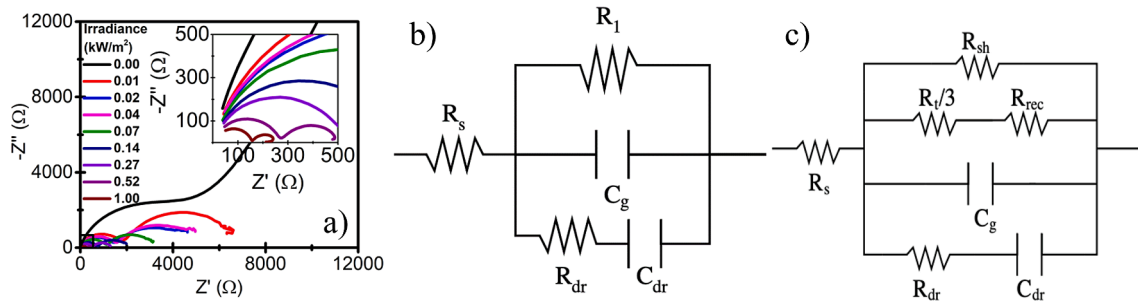


Fig. 5. A) representative nyquist plot of the sample 0% mabr at day 15, with zoom of the high frequency region (inset). b) Simplified and c) extended version of the equivalent circuit of perovskite solar cells used for impedance fitting.

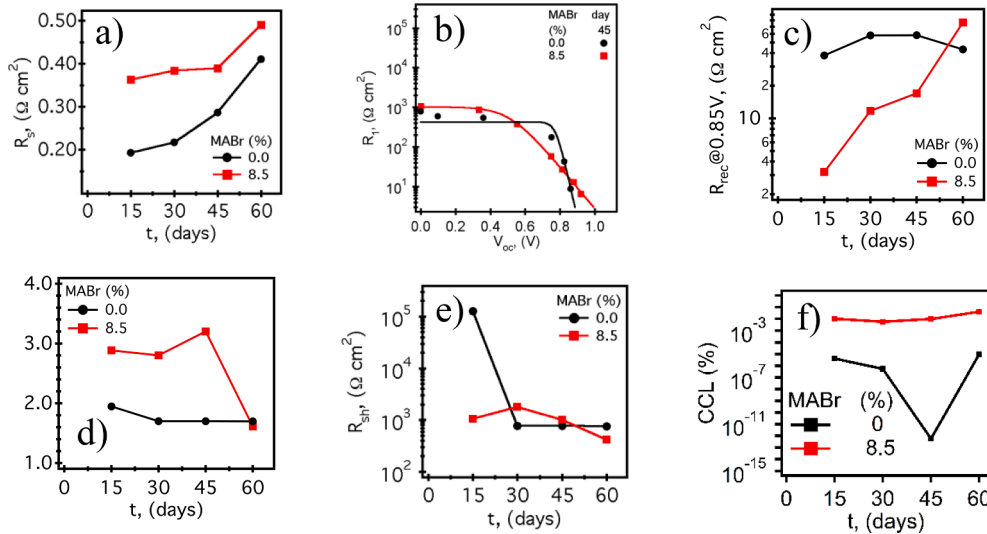


Fig. 6. Resistance parameters obtained by the EIS fitting of both samples. a) Temporal evolution of R_s . b) R_1 at day 45. c) Temporal evolution of R_{rec} at 0.85 V. d) Temporal evolution of ideality factor m . e) Temporal evolution of R_{sh} . f) Temporal evolution of CCL.

the measurements were made in open-circuit conditions and perovskite have very high charge carriers mobility, R_t is negligible [48]. This behavior is described with the diode equation (Eqs (1) and (2) [47,60]:

$$\frac{1}{R_1} = \frac{1}{R_{rec}} + \frac{1}{R_{sh}} \quad (1)$$

$$R_{rec} = R_{rec,0} e^{-\frac{qV}{mk_B T}} \quad (2)$$

where $R_{rec,0}$ is the recombination resistance at $V = 0$, q is the charge of an electron, k_B is the Boltzmann constant, T is the temperature and m is the ideality factor. Since m was calculated from this model, it represents the effective or apparent ideality factor, which considers the ionic movement and electron dynamics [48]. With this equation, it is possible to fit the data (solid line in Fig. 6b) to determine the value for R_{rec} as function of the voltage and R_{sh} .

Using the fitting results, punctual values for R_{rec} at 0.85 V and R_{sh} were determined for both compositions during the 60 days period of aging (Fig. 6c and Fig. 6e respectively), with a maximum uncertainty percentage of $\pm 10\%$. These values are unique within the error range. The 0% MABr cells present almost the same R_{rec} throughout the aging period, which means that in these samples, recombination processes barely changes with time. Meanwhile, the 8.5% MABr samples present a progressively increasing R_{rec} as time passes, starting with a lower value than 0% MABr (at $3.13 \Omega \text{ cm}^2$) and finishing with a higher value than 0% MABr (at $62.68 \Omega \text{ cm}^2$). These results suggests that, during the aging process, the defects in the 8.5% MABr PSCs are reduced (probably due to recrystallization processes), resulting in a reduction in recombination

phenomena.

Analyzing the R_{sh} behavior, it is observed that 0% MABr cells possess a high value (over $1 \times 10^5 \Omega \text{ cm}^2$) at Day 15, but suffers a massive loss on the following days, indicating the formation of imperfections on the layers of the PSCs, promoting new pathways for charge leakages. In contrast, the cells containing MABr presents relatively low value at Day 15, meaning more leaks than undoped cells on the same day, then suffers a slight increase at Day 30 (slightly less leaks than Day 15), and finally, presents a decrease on the rest of the aging period, reflecting a deterioration in the perovskite layer that promoted more leaks as time passed on.

Fig. 6d shows the ideality factor m of the samples. This parameter indicates the predominant type of recombination present in the solar cells. For the samples without MABr, m is constant across the aging period, presenting a value of 2, implying that those cells the Shockley-Reed-Hall (non-radiative) recombination process is dominant [9,49,51,54]. Meanwhile, the 8.5% MABr samples present “ m ” values around 3 on the first 45 days, which indicates the presence of more complex recombination processes on multiple energy levels attributed to the formation of a huge amount of defects [9,51], and then decreases to 2 on day 60, indicating a passivation of those defects during degradation on day 60.

3.4.2. High frequency resistance (R_{HF})

R_{rec} provides information about the effective recombination processes, showing both the ionic and electronic influences, which are the effective phenomena that the solar cell yields during operation. Using

R_{rec} and R_{dr} , it is possible to calculate the high frequency resistance (R_{HF}), which gives insight only on the electronic effects on the recombination processes [37]. R_{HF} can be calculated using Eq. (3) [57]:

$$R_{HF} = \frac{(R_1) \bullet R_{dr}}{(R_1) + R_{dr}} \quad (3)$$

The graphics depicting R_{HF} at different voltages are summarized in Fig. S4. Those graphs depict a similar behavior to the ones for R_1 and R_{rec} , with their values being a magnitude order lower, meaning that the ionic processes have a strong effect on the solar cells.

From R_{HF} , it is possible to determine the charge collection loss (CCL) at short-circuit conditions using Eqs. (4), 5 and 6 [37,47,60]:

$$CCL = \left(\frac{J_{rec}(V_{SC}) + J_{sh}(V_{SC})}{J_{rec}(V_{OC}) + J_{sh}(V_{OC})} \right) \times 100 \quad (4)$$

$$J_{rec} = \frac{mk_B T}{qR_{rec}} \quad (5)$$

$$J_{sh} = \frac{V}{R_{sh}} \quad (6)$$

where $J_{sh}(V_{SC})$ and $J_{rec}(V_{SC})$ are the Shunt and recombination current at short circuit and $J_{sh}(V_{OC})$ $J_{rec}(V_{OC})$ are the same currents evaluated at open circuit voltage.

Fig. 6f shows the temporal evolution of CCL. For both compositions, charge losses are basically inexistant across the aging period, being slightly lower in the 0% MABr PSCs. Since this parameter is strongly related to the internal quantum efficiency (IQE), these results suggest that the changes observed on EQE (Fig. 4) are mainly attributed to changes in the light absorption processes, since:

$$EQE = \eta_{lhc} \cdot IQE \quad (7)$$

where η_{lhc} is the absorptivity (or light harvesting efficiency) [7].

3.4.3. Ionic components

The last parameters provided by the used equivalent circuit are R_{dr} and C_{dr} , which give information about the amount of ionic vacancies and ionic mobility.

8.5% MABr samples at day 15 possess lower R_{dr} (Fig. 7a) than the 0% MABr samples, which could mean that 8.5% MABr PSCs present either a higher amount of vacancies or a higher ionic mobility. At the same time, C_{dr} (Fig. 7b) is lower in the 8.5% MABr cells compared to the cells without MABr, corroborating that the effect in R_{dr} is due to a higher ionic mobility in the cells with MABr [37]. This trend is followed across the first 45 days of the aging period (Fig. 7c y d, Fig. S5 and Fig. S6). Analyzing the temporal evolution 0% MABr cells it is observed that R_{dr} and C_{dr} remained almost constant from day 15 to day 45. At day 60 C_{dr} presented a notorious increase accompanied with a small increase in R_{dr} , suggesting a reduction in the ion mobility at this final time. On the other hand, the 8.5% MABr cells presented a R_{dr} increase from day 15 to day 45 mean while C_{dr} decreases in the same period suggesting a reduction in the ion vacancies of these samples. Whoever, on day 60 the trend is reverted with a reduction of R_{dr} and a considerable increase of C_{dr} , indicating a high formation of ionic vacancies for on 8.5% MABr PSCs at the end of the ageing study.

Ionic diffusion/movement can be related to the formation of segregates, which explains the presence Cs phases (according to IPCE) on 8.5% MABr PSCs on the first 45 days, while ionic vacancies are related to the formation of degradation products, which further confirms the inevitable degradation that both samples suffered at day 60 [37].

4. Conclusions

Two types of CsFAPbI₃ solar cells were prepared: one type with MABr doping, and one without doping.

The addition of MABr to the CsFAPbI₃ matrix improved the stability of the photovoltaic performance under highly humid conditions. The MABr doping helped for the PSCs to have better photoabsorption, as it is shown by the IPCE measurements. These measurements suggests that

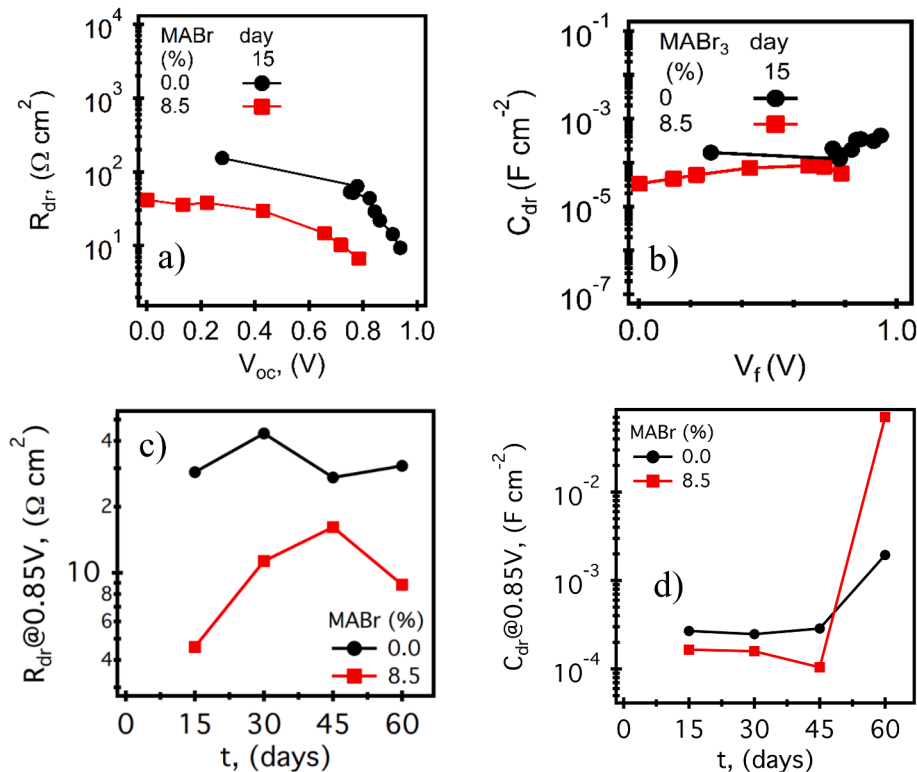


Fig. 7. a) R_{dr} at day 15, b) C_{dr} at day 15, and temporal evolution of c) R_{dr} and d) C_{dr} at 0.85 V.

the cause of a better photoabsorption is due to the formation of photoactive phases, which allowed the devices to absorb more photons near the infrared range.

EIS studies successfully provided data on the electronic properties of the PSCs, the different mechanisms of the charge carriers transport and how they changed during the aging period. These studies showed that the MABr-less samples have a better electronic performance, as shown by higher recombination resistances and lower ideality factor (meaning less trap states). But even if the 0% MABr cells had the better electronic processes from the two types of samples, the ionic data provided by EIS indicates that the 8.5% MABr devices have the highest ionic diffusion from the two compositions, reason why these cells present the formation of photoactive phases, which grant a better long-term efficiency compared to the 0% MABr samples.

This paper demonstrates the utility of EIS to understand the electrical and ionic dynamics of the PSCs, and how it is useful in the design of new technologies.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.solener.2023.04.058>.

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